Capstone Checkpoint 4.1 - VALIDATION SET - FINAL

Oracle, vector, and hybrid experiment comparison. Generated 2026-09-15 19:41 EDT.

15	4	375	25	0
Included runs	Configurations	Question-runs	Questions	Legacy runs excluded

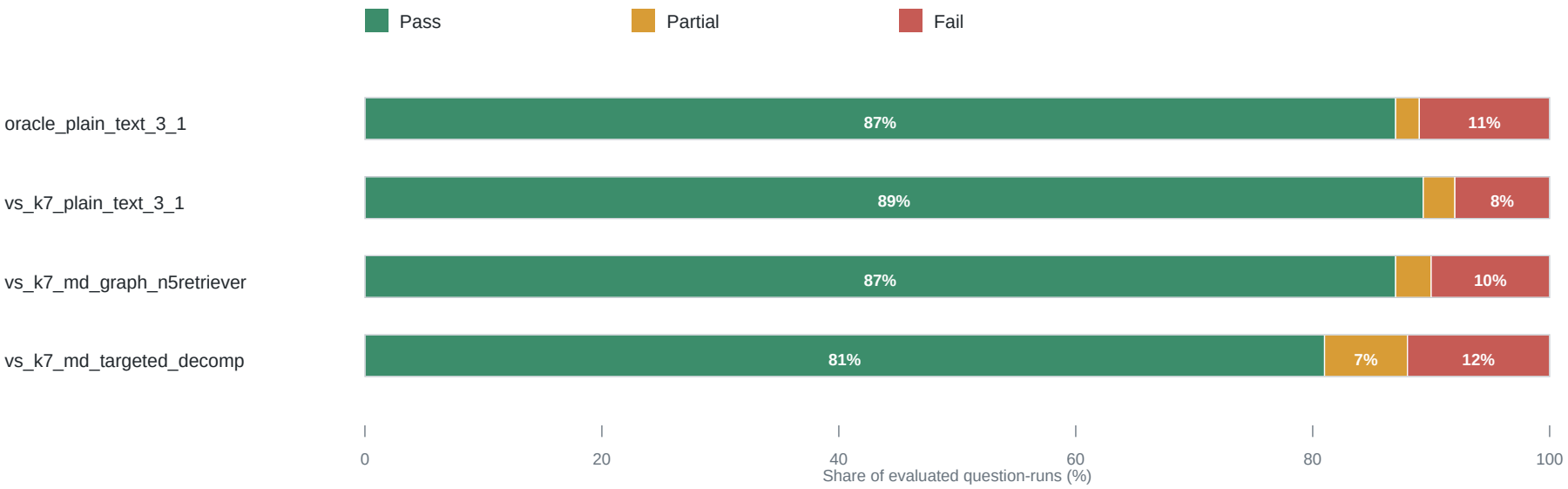
Summary

The highest observed mean quality score was 0.907 for vs_k7_plain_text_3_1 across 3 run(s). Quality assigns 1 to pass, 0.5 to partial, and 0 to fail.

The highest observed mean designated-source recall was 97.3% for vs_k7_plain_text_3_1. Article recall is recomputed from the required and retrieved article names in every CSV.

The highest task-family attention score was 40.0 for Exhaustive list or time filter under vs_k7_plain_text_3_1. Attention is 100 x (1 - grouped quality), so higher values identify larger observed improvement opportunities.

Answer outcomes



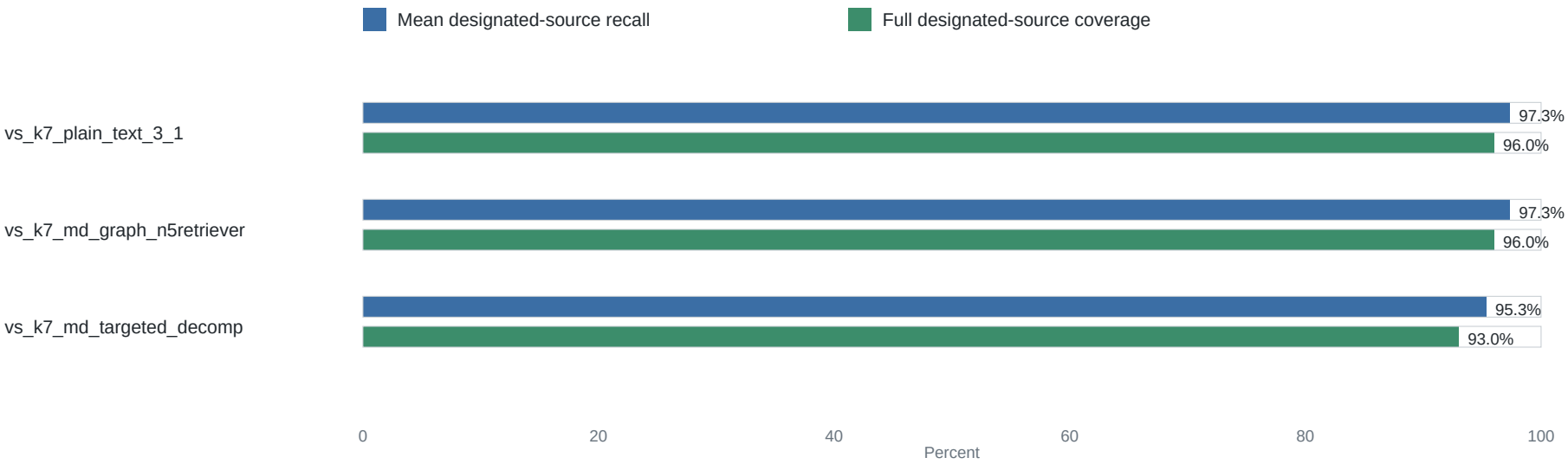
Configuration	Runs	Cases	Pass	Partial	Fail	Quality	Run SD	Article recall	Full source coverage	Mean context
oracle_plain_text_3_1	4	100	87.0%	2.0%	11.0%	0.880	0.032	n/a	n/a	58.8k
vs_k7_plain_text_3_1	3	75	89.3%	2.7%	8.0%	0.907	0.019	97.3%	96.0%	130.9k
vs_k7_md_graph_n5retriever	4	100	87.0%	3.0%	10.0%	0.885	0.030	97.3%	96.0%	237.7k

Configuration	Runs	Cases	Pass	Partial	Fail	Quality	Run SD	Article recall	Full source coverage	Mean context
vs_k7_md_targeted_decomp	4	100	81.0%	7.0%	12.0%	0.845	0.022	95.3%	93.0%	175.1k

Percentages aggregate all question-runs within a configuration. Run SD is the standard deviation of mean quality across repeated runs.

Retrieval diagnostics

Designated-source coverage



The analyzer recomputes recall after removing only final .txt and .html suffixes. This corrects historical .html.txt versus .html mismatches. Designated-source recall is diagnostic: alternative retrieved articles may still contain sufficient evidence.

Outcome and source-coverage combinations

Configuration	Pass, full	Pass, incomplete	Non-pass, full	Non-pass, missing
vs_k7_md_graph_n5retriever	87	0	9	4
vs_k7_md_targeted_decomp	80	1	13	6
vs_k7_plain_text_3_1	67	0	5	3

A non-pass result with full designated-source recall points toward answer synthesis, context noise, table interpretation, or judge behavior. A non-pass result with a missing designated source is consistent with a retrieval recall problem, but manual review is still required.

Per-question comparison

Cells show the mean quality score for each question and configuration across repeated runs.

	oracle_plain_text_3_1	vs_k7_plain_text_3_1	vs_k7_md_graph_n5retriever	vs_k7_md_targeted_decomp
VAL01	**1.00**	**1.00**	**1.00**	**1.00**
VAL02	**0.00**	**0.00**	**0.00**	**0.00**
VAL03	**1.00**	**1.00**	**1.00**	**1.00**
VAL04	**1.00**	**1.00**	**1.00**	**0.75**
VAL05	**0.12**	**1.00**	**0.50**	**0.38**
VAL06	**0.75**	**1.00**	**1.00**	**0.75**
VAL07	**0.25**	**0.00**	**0.00**	**0.00**
VAL08	**1.00**	**1.00**	**0.88**	**0.62**
VAL09	**1.00**	**1.00**	**1.00**	**0.75**
VAL10	**1.00**	**0.83**	**0.88**	**1.00**
VAL11	**1.00**	**1.00**	**1.00**	**1.00**
VAL12	**1.00**	**1.00**	**1.00**	**1.00**
VAL13	**1.00**	**1.00**	**1.00**	**1.00**
VAL14	**0.88**	**0.83**	**0.88**	**0.88**
VAL15	**1.00**	**1.00**	**1.00**	**1.00**
VAL16	**1.00**	**1.00**	**1.00**	**1.00**
VAL17	**1.00**	**1.00**	**1.00**	**1.00**
VAL18	**1.00**	**1.00**	**1.00**	**1.00**
VAL19	**1.00**	**1.00**	**1.00**	**1.00**
VAL20	**1.00**	**1.00**	**1.00**	**1.00**
VAL21	**1.00**	**1.00**	**1.00**	**1.00**
VAL22	**1.00**	**1.00**	**1.00**	**1.00**
VAL23	**1.00**	**1.00**	**1.00**	**1.00**
VAL24	**1.00**	**1.00**	**1.00**	**1.00**
VAL25	**1.00**	**1.00**	**1.00**	**1.00**

Fail (0) Partial (0.5) Pass (1)

Question	oracle_plain_text_3_1	vs_k7_plain_text_3_1	vs_k7_md_graph_n5retriever	vs_k7_md_targeted_decomp
VAL01	1.000	1.000	1.000	1.000
VAL02	0.000	0.000	0.000	0.000
VAL03	1.000	1.000	1.000	1.000
VAL04	1.000	1.000	1.000	0.750
VAL05	0.125	1.000	0.500	0.375
VAL06	0.750	1.000	1.000	0.750
VAL07	0.250	0.000	0.000	0.000
VAL08	1.000	1.000	0.875	0.625
VAL09	1.000	1.000	1.000	0.750
VAL10	1.000	0.833	0.875	1.000
VAL11	1.000	1.000	1.000	1.000
VAL12	1.000	1.000	1.000	1.000
VAL13	1.000	1.000	1.000	1.000
VAL14	0.875	0.833	0.875	0.875
VAL15	1.000	1.000	1.000	1.000
VAL16	1.000	1.000	1.000	1.000
VAL17	1.000	1.000	1.000	1.000
VAL18	1.000	1.000	1.000	1.000
VAL19	1.000	1.000	1.000	1.000
VAL20	1.000	1.000	1.000	1.000
VAL21	1.000	1.000	1.000	1.000
VAL22	1.000	1.000	1.000	1.000
VAL23	1.000	1.000	1.000	1.000
VAL24	1.000	1.000	1.000	1.000
VAL25	1.000	1.000	1.000	1.000

Per-question article recall

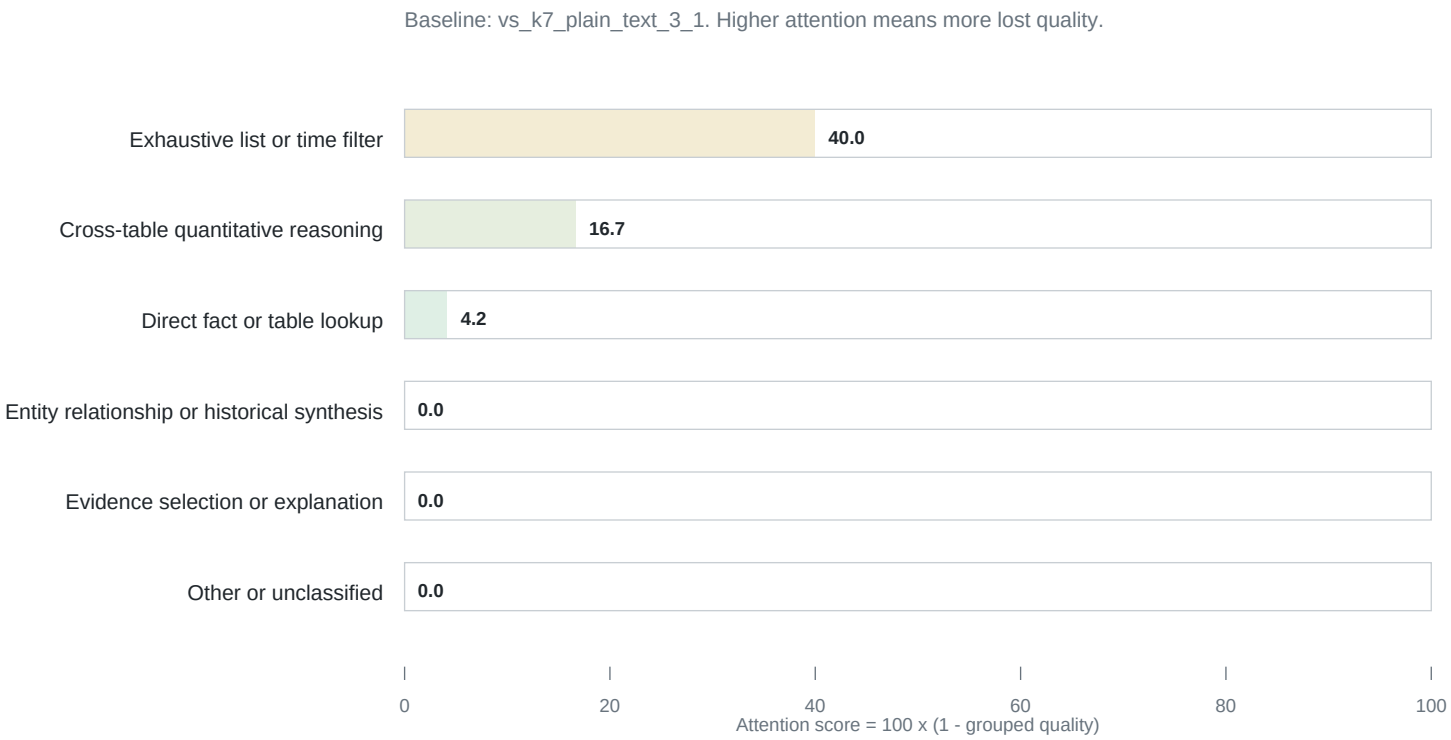
Fraction of each question's designated source articles that appeared in the retrieved set (retriever configurations only). 1.00 = every gold article retrieved; lower values mark a retrieval miss — usually a multi-source question.

Question	vs_k7_plain_text_3_1	vs_k7_md_graph_n5retriever	vs_k7_md_targeted_decomp
VAL01	100.0%	100.0%	100.0%
VAL02	33.3%	33.3%	33.3%
VAL03	100.0%	100.0%	100.0%
VAL04	100.0%	100.0%	75.0%
VAL05	100.0%	100.0%	100.0%
VAL06	100.0%	100.0%	100.0%
VAL07	100.0%	100.0%	100.0%
VAL08	100.0%	100.0%	100.0%
VAL09	100.0%	100.0%	75.0%
VAL10	100.0%	100.0%	100.0%
VAL11	100.0%	100.0%	100.0%
VAL12	100.0%	100.0%	100.0%
VAL13	100.0%	100.0%	100.0%
VAL14	100.0%	100.0%	100.0%
VAL15	100.0%	100.0%	100.0%
VAL16	100.0%	100.0%	100.0%
VAL17	100.0%	100.0%	100.0%
VAL18	100.0%	100.0%	100.0%
VAL19	100.0%	100.0%	100.0%
VAL20	100.0%	100.0%	100.0%
VAL21	100.0%	100.0%	100.0%

Question	vs_k7_plain_text_3_1	vs_k7_md_graph_n5retriever	vs_k7_md_targeted_decomp
VAL22	100.0%	100.0%	100.0%
VAL23	100.0%	100.0%	100.0%
VAL24	100.0%	100.0%	100.0%
VAL25	100.0%	100.0%	100.0%

Performance by task type

The 18 original category labels are consolidated into five task families so each group contains multiple questions. The attention score ranks observed improvement opportunities for the strongest retrieval configuration; it is not a statistical significance test.



Task family	Questions	Baseline quality	Attention	Fail	Oracle quality	Oracle gap	Article recall
Exhaustive list or time filter	5	0.600	40.0	40.0%	0.475	-0.125	86.7%
Cross-table quantitative reasoning	1	0.833	16.7	0.0%	1.000	+0.167	100.0%
Direct fact or table lookup	4	0.958	4.2	0.0%	0.969	+0.010	100.0%
Entity relationship or historical synthesis	1	1.000	0.0	0.0%	1.000	+0.000	100.0%
Evidence selection or explanation	10	1.000	0.0	0.0%	0.975	-0.025	100.0%
Other or unclassified	4	1.000	0.0	0.0%	1.000	+0.000	100.0%

Baseline quality and failure rate use vs_k7_plain_text_3_1. Oracle gap equals oracle quality minus baseline quality: a large positive gap is consistent with a retrieval or context-selection opportunity, while low oracle quality indicates that the difficulty persists even when the designated articles are supplied.

Performance by source scope

Source scope groups questions by the number of designated articles. It tests whether the number of sources, rather than the reasoning operation, explains performance.

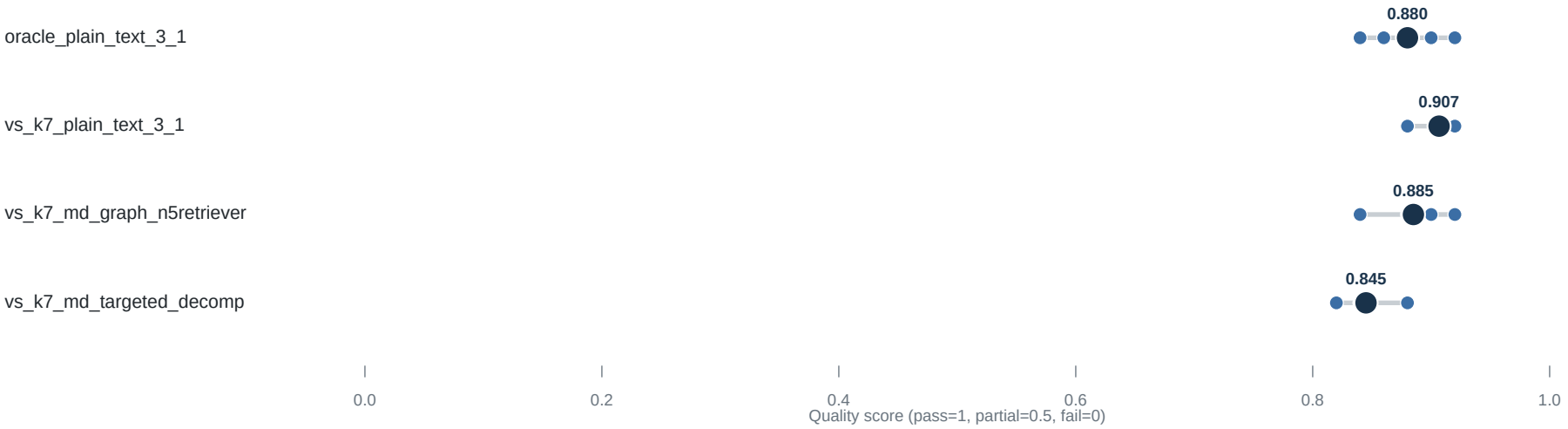
	oracle_plain_text_3_1	vs_k7_plain_text_3_1	vs_k7_md_graph_n5retriever	vs_k7_md_targeted_decomp
1 article	**0.889**	**0.935**	**0.903**	**0.854**
2 articles	**1.000**	**0.972**	**0.979**	**0.958**
3+ articles	**0.000**	**0.000**	**0.000**	**0.000**

Source scope	Configuration	Questions	Question-runs	Pass	Partial	Fail	Quality	Article recall
1 article	oracle_plain_text_3_1	18	72	87.5%	2.8%	9.7%	0.889	n/a
2 articles	oracle_plain_text_3_1	6	24	100.0%	0.0%	0.0%	1.000	n/a
3+ articles	oracle_plain_text_3_1	1	4	0.0%	0.0%	100.0%	0.000	n/a
1 article	vs_k7_plain_text_3_1	18	54	92.6%	1.9%	5.6%	0.935	100.0%
2 articles	vs_k7_plain_text_3_1	6	18	94.4%	5.6%	0.0%	0.972	100.0%
3+ articles	vs_k7_plain_text_3_1	1	3	0.0%	0.0%	100.0%	0.000	33.3%
1 article	vs_k7_md_graph_n5retriever	18	72	88.9%	2.8%	8.3%	0.903	100.0%
2 articles	vs_k7_md_graph_n5retriever	6	24	95.8%	4.2%	0.0%	0.979	100.0%
3+ articles	vs_k7_md_graph_n5retriever	1	4	0.0%	0.0%	100.0%	0.000	33.3%
1 article	vs_k7_md_targeted_decomp	18	72	80.6%	9.7%	9.7%	0.854	98.6%
2 articles	vs_k7_md_targeted_decomp	6	24	95.8%	0.0%	4.2%	0.958	95.8%
3+ articles	vs_k7_md_targeted_decomp	1	4	0.0%	0.0%	100.0%	0.000	33.3%

Question-runs are repeated measurements of the listed unique questions. Source-scope groups remain small and mix different task families, so this view should be used as a diagnostic rather than a general performance estimate.

Run-to-run variability

Each dot is one run; the larger dot is the configuration mean.



Run	Configuration	Cases	Pass	Partial	Fail	Quality	Article recall	Full source coverage
bold_rivest	oracle_plain_text_3_1	25	21	1	3	0.860	n/a	n/a
confident_ullman	oracle_plain_text_3_1	25	23	0	2	0.920	n/a	n/a
epic_mccarthy	oracle_plain_text_3_1	25	22	1	2	0.900	n/a	n/a
quirky_kernighan	oracle_plain_text_3_1	25	21	0	4	0.840	n/a	n/a
focused_knuth	vs_k7_plain_text_3_1	25	23	0	2	0.920	97.3%	96.0%
quizzical_goldwasser	vs_k7_plain_text_3_1	25	23	0	2	0.920	97.3%	96.0%
romantic_brin	vs_k7_plain_text_3_1	25	21	2	2	0.880	97.3%	96.0%
bold_valiant	vs_k7_md_graph_n5retriever	25	22	1	2	0.900	97.3%	96.0%
elegant_matsumoto	vs_k7_md_graph_n5retriever	25	22	0	3	0.880	97.3%	96.0%
thirsty_turing	vs_k7_md_graph_n5retriever	25	20	2	3	0.840	97.3%	96.0%
elegant_backus	vs_k7_md_graph_n5retriever	25	23	0	2	0.920	97.3%	96.0%
practical_thompson	vs_k7_md_targeted_decomp	25	19	3	3	0.820	95.3%	92.0%

Run	Configuration	Cases	Pass	Partial	Fail	Quality	Article recall	Full source coverage
focused_tarjan	vs_k7_md_targeted_decomp	25	21	2	2	0.880	97.3%	96.0%
optimistic_tarjan	vs_k7_md_targeted_decomp	25	21	0	4	0.840	97.3%	96.0%
musing_matsumoto	vs_k7_md_targeted_decomp	25	20	2	3	0.840	91.3%	88.0%

Methodology and limitations

Scoring

Pass is assigned 1, partial is assigned 0.5, and fail is assigned 0. The quality score is the arithmetic mean across the applicable question-runs. Pass, partial, and fail percentages are also reported separately so the weighted score does not hide the underlying verdicts.

Task-family attention

The attention score is 100 multiplied by one minus grouped quality. A score of 66 means that the group lost 66% of the available quality points under the selected retrieval baseline. It is a prioritization aid, not an estimate of population-wide failure risk. Task-family and source-scope results must be read with their unique-question counts because the evaluation set contains only 19 questions.

Configuration grouping

Runs are grouped by a CSV experiment_name or config_name when one is present. If no explicit name is present, grouping falls back to evaluation mode, context source, retriever kind and label, Top K, hybrid weights, answer model, and judge model. RAGAS-generated filenames identify individual runs but do not define configurations.

Framework compatibility

Legacy oracle runs stored under pass_fail_only directories are excluded by default because they could not produce partial verdicts. Use --include-binary-oracle only for a labeled historical appendix, not for a direct three-level score comparison.

Interpretation limits

An LLM produced the answers and another LLM applied the rubric, so repeated runs can vary. Article recall measures coverage of designated ground-truth sources, not every possible supporting source. Full article recall does not prove that the model used the evidence correctly, and missing a designated source does not prove the context was insufficient when another article contains equivalent evidence. Differences between configurations should be interpreted together with run counts and per-question behavior.

Included experiment files

Run file
experiments/bold_rivest.csv
experiments/bold_valiant.csv
experiments/confident_ullman.csv
experiments/elegant_backus.csv
experiments/elegant_matsumoto.csv
experiments/epic_mccarthy.csv
experiments/focused_knuth.csv
experiments/focused_tarjan.csv
experiments/musing_matsumoto.csv
experiments/optimistic_tarjan.csv
experiments/practical_thompson.csv

Run file
experiments/quirky_kernighan.csv
experiments/quizzical_goldwasser.csv
experiments/romantic_brin.csv
experiments/thirsty_turing.csv

Excluded files

Run file	Reason
datasets/wikipedia_GT_validation_set_advanced_inputs.csv	missing columns: ['score']
datasets/wikipedia_GT_validation_set_graph_inputs.csv	duplicate of datasets/wikipedia_GT_validation_set_advanced_inputs.csv
datasets/wikipedia_GT_validation_set_oracle_inputs.csv	missing columns: ['score']
datasets/wikipedia_GT_validation_set_retrieval_inputs.csv	duplicate of datasets/wikipedia_GT_validation_set_oracle_inputs.csv

Warnings

Task-family analysis left these categories unclassified: Cross-article comparison, Diagnostic reasoning and limitations, Inference limitation, People connection: shared work and roles